

A 31-Chip Counterexample to the Full-Subtree Ballot Conjecture for Labeled Chip-Firing on Binary Trees

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

September 11, 2026

Abstract

In labeled chip-firing on the infinite rooted ordered binary tree, start the chips labeled $1, \dots, 2^\ell - 1$ at the root. A conjecture of Inagaki, Khovanova, and Luo asserts that, in every reachable stable configuration, the ordered labels in the complete left descendant subtree of any vertex are termwise smaller than those in its right descendant subtree. We disprove the conjecture at $\ell = 5$. An explicit sequence of 72 legal firings reaches a stable configuration of the 31 chips for which, at the root, the seventh left and right order statistics are respectively 14 and 13. The full firing trace and terminal labeling are included, so reachability and the failed inequality can be checked using only integer comparisons and pile updates.

1 The dynamics and the conjecture

Index the vertices of the infinite rooted ordered binary tree by the nonnegative integers. The root is 0, the children of v are

$$L(v) = 2v + 1, \quad R(v) = 2v + 2,$$

and the parent map is

$$p(0) = 0, \quad p(v) = \left\lfloor \frac{v-1}{2} \right\rfloor \quad (v > 0).$$

If a vertex contains three chips with distinct labels $x < y < z$, a legal firing may select them and send x to $L(v)$, y to $p(v)$, and z to $R(v)$. Thus the middle chip selected at the root remains at the root. A configuration is stable when every vertex contains at most two chips.

For an integer $\ell \geq 1$, place the chips labeled $1, \dots, 2^\ell - 1$ at the root. In a reachable stable configuration, let

$$a_1 < \dots < a_m, \quad b_1 < \dots < b_m, \quad m = 2^{\ell-d-1} - 1,$$

be the labels in the complete descendant subtrees rooted at $L(v)$ and $R(v)$, respectively, where v has depth $d \leq \ell - 2$. Inagaki, Khovanova, and Luo conjectured that

$$a_i < b_i \quad (1 \leq i \leq m) \tag{1}$$

for every ℓ , every reachable stable configuration, and every eligible vertex v [1, Conjecture 3.7]. They report an exhaustive check of the 36,220 stable configurations on 15 chips. The witness below occurs at the next parameter value, 31 chips; no minimality claim beyond the reported 15-chip computation is needed for the disproof.

2 An exact reachable counterexample

Each entry $(v; x, y, z)$ in Table 1 means: fire vertex v using the three chips $x < y < z$. The step numbers are read down each of the three two-column blocks.

Theorem 1. *Starting with labels $1, \dots, 31$ at the root, the 72 firings in Table 1 are legal and terminate in a stable configuration. At the root of that configuration, $a_7 = 14$ and $b_7 = 13$. Consequently, (1) is false.*

Proof. Begin with the pile $P_0 = \{1, \dots, 31\}$ and all other piles empty. For each successive table entry $(v; x, y, z)$, remove x, y, z from P_v and insert them into $P_{L(v)}, P_{p(v)}, P_{R(v)}$, respectively. Direct induction through the table verifies that the named three labels belong to P_v before every step; their displayed order also verifies $x < y < z$. Thus the table is a finite certificate of legal reachability under exactly the rule above.

After step 72, the chip at each heap vertex $0, 1, \dots, 30$, in order, is

$$(22, 16, 23, 7, 21, 12, 26, 2, 10, 15, 24, 6, 13, 17, 30, \\ 1, 4, 5, 19, 14, 20, 18, 28, 3, 11, 8, 29, 9, 27, 25, 31).$$

Every label from 1 through 31 appears exactly once in this list. Since a firing preserves the label multiset, no chip lies elsewhere. Each occupied pile therefore has size one, so the endpoint is stable.

The labels in the complete left and right child subtrees of the root, sorted in increasing order, are

$$A = (1, 2, 4, 5, 7, 10, 14, 15, 16, 18, 19, 20, 21, 24, 28), \\ B = (3, 6, 8, 9, 11, 12, 13, 17, 23, 25, 26, 27, 29, 30, 31).$$

Both lists have the required length $2^{5-1} - 1 = 15$, but their seventh entries satisfy $a_7 = 14 > 13 = b_7$. Hence the universally quantified strict inequality fails. $\square$

3 Verification, context, and limitations

The certificate requires neither a termination theorem nor an assumption about other firing schedules: a single reachable stable endpoint refutes the universal statement. A deterministic replay checks, after each row, strict label order, membership of all three selected chips in the fired pile, the left–parent–right update, and preservation of the complete label multiset. A separately implemented replay represents the state by chip positions rather than vertex piles and gives the same terminal labeling and subtree lists. The complete table also permits a verifier to be reimplemented directly from the definition without trusting the search procedure.

The original paper’s positive structural results are not challenged here. This note only negates the full-subtree ballot assertion; it does not classify the reachable stable configurations for 31 chips, determine which weaker ballot inequalities survive, or independently reproduce the reported 15-chip census. For broad adjacent methodology, exact certificate and independent label-verifier techniques also occur in a tree-labeling setting in [2]; that work is not about chip-firing and supplies no mathematical input to Theorem 1.

Computational and AI assistance. The trace was found by an exact event-order search using a threshold coloring, then lifted to distinct labels. Deterministic Python programs performed the two exact replays described above; floating-point arithmetic was not used. AI language models assisted with search design, code drafting, a separate verification pass, and manuscript preparation. The disproof itself depends only on the explicit finite trace and the deterministic chip-firing rule.

Table 1: Complete labeled firing certificate. Each quadruple is $(v; x, y, z)$, with v the fired heap vertex and $x < y < z$ the selected labels.

Step	Firing	Step	Firing	Step	Firing
1	0; 1, 2, 3	25	0; 20, 24, 25	49	2; 11, 26, 30
2	0; 14, 15, 16	26	0; 24, 26, 27	50	3; 4, 10, 19
3	0; 2, 15, 17	27	0; 26, 28, 29	51	4; 20, 24, 28
4	0; 4, 5, 6	28	0; 28, 30, 31	52	5; 11, 22, 29
5	0; 5, 7, 8	29	2; 22, 23, 25	53	6; 9, 13, 30
6	1; 1, 2, 14	30	2; 11, 12, 13	54	0; 16, 23, 26
7	0; 15, 18, 19	31	5; 3, 6, 8	55	1; 10, 16, 24
8	0; 2, 7, 9	32	1; 19, 20, 24	56	2; 13, 22, 26
9	0; 7, 10, 11	33	0; 20, 23, 30	57	7; 1, 2, 4
10	0; 10, 12, 13	34	6; 17, 21, 25	58	8; 5, 7, 19
11	0; 18, 20, 21	35	0; 7, 10, 12	59	9; 14, 15, 20
12	1; 2, 4, 15	36	2; 6, 21, 27	60	0; 16, 22, 23
13	1; 5, 7, 18	37	2; 29, 30, 31	61	10; 18, 21, 28
14	2; 3, 16, 17	38	0; 21, 23, 30	62	4; 15, 21, 24
15	2; 6, 8, 9	39	1; 7, 20, 21	63	11; 3, 6, 11
16	0; 4, 7, 8	40	3; 2, 4, 7	64	12; 8, 12, 29
17	0; 16, 20, 22	41	1; 4, 26, 28	65	5; 6, 12, 13
18	2; 8, 19, 21	42	0; 20, 23, 26	66	13; 9, 17, 27
19	3; 1, 2, 5	43	2; 12, 26, 30	67	14; 25, 30, 31
20	4; 14, 15, 18	44	6; 27, 30, 31	68	6; 17, 26, 30
21	1; 2, 4, 15	45	5; 6, 11, 12	69	2; 12, 23, 26
22	0; 4, 7, 12	46	4; 15, 16, 21	70	3; 2, 7, 10
23	1; 4, 10, 16	47	0; 10, 23, 26	71	1; 7, 16, 21
24	0; 19, 20, 23	48	1; 10, 16, 20	72	0; 16, 22, 23

Data availability. The machine-readable 72-step trace and deterministic replay programs accompany the project materials. Table 1 contains the complete decisive certificate.

References

- [1] R. Inagaki, T. Khovanova, and A. Luo, *On Chip-Firing on Undirected Binary Trees*, arXiv:2410.00039v1, 2024. <https://arxiv.org/abs/2410.00039>.
- [2] Z. Zeng, *Odd-Order Trees with Exactly Three Vertices of Degree Two Are Edge-Graceful*, SSRN preprint 7387319, 2026. <https://doi.org/10.2139/ssrn.7387319>.